F20DL GROUP 16

Rice variety classification: final results

Draft pack for group review · 11 Nov 2026

99.92%

HEADLINE RESULT

Best test accuracy (neural network on 106 measurements, mean of 3 seeds)

11,206 unseen test grains, each test read spent once

DATASET

74,703

grains, 5 varieties (75,000 images, 297 exact duplicates removed)

RUN ORDER

16 of 16

run-order steps passed

CHECKS

439

automated tests passed

TEST READS

3 of 3

test reads used, once each, on one frozen build

TEST ACCURACY BY MODEL

Neural network (MLP)	99.92%
Gradient boosting 13 errors	99.88%
Logistic regression	99.83%
k-nearest neighbours	99.79%
Image CNN	99.69%
Naive Bayes	99.05%

KNOWS WHEN IT DOES NOT KNOW

10,979 answered

The routing chain answered 98.0% of grains with 2 errors. It flagged 175 grains as unfamiliar instead of guessing.

HONEST LIMITS

- Jasmine from a batch never seen in training: 1.2% recall, the model is confidently wrong.
- The safety chain flagged 3,272 of 3,283 of those grains as unfamiliar.
- A colour-balance change drops the image CNN from 100% to 2-7%.

WHAT IS IN THE PACK

- Six-page report (PDF)
- Speaker plan and Q&A guide
- 13 disclosures, 33-file manifest
- Five-slide viva deck (PPTX + PDF)
- README and submission checklist

STILL OPEN FOR THE GROUP

Names and contributions · viva speakers · AI-use statement (brief section 8)

DUE

Wed 25 Nov 2026, 15:30

github.com/yahiaelmahdi7-bit/f20dl-rice-classification